

Chandana

Azure Data Engineer

PROFESSIONAL SUMMARY:

- 8+ years of IT experience specializing in Database Design, Retrieval, Manipulation and Support Requirements Analysis, Application Development, Testing, Implementation and Deployments using MS SQL Server 2019/2016, Oracle **PL/SQL** 19g/11g, **MSBI**.
- Experience in **Data Center Migration**, Azure Data Factory (ADF) V2. Managing Database, Azure Data Platform services (Azure Data Lake (**ADLS**), Data Factory (**ADF**), Data Lake Analytics, **Stream Analytics**, Azure SQL DW, **HDInsight/Databricks**, **NoSQL** DB), **SQL Server**, Oracle, Data Warehouse etc.
- Experience in **SDLC** (Software Development Life Cycle) involving System analysis, Design Development & Implementation using **Waterfall methodology** and **agile methodology**.
- Experience with Azure transformation projects and implementing **ETL** and data movement solutions using Azure Data Factory (ADF), SSIS.
- Involved in building an information pipeline and performed analysis utilizing AWS stack (EMR, EC2, S3, RDS, Lambda, Glue, SQS, Redshift, Snowflake), collaborated with multi-functional roles to communicate and align the development efforts.
- Recreating existing application logic and functionality in the **Azure Data Lake**, **Data Factory**, SQL Database and SQL Datawarehouse environment.
- Implement ad-hoc analysis solutions using Azure Data Lake Analytics/Store, HDInsight.
- Design & implement migration strategies for traditional systems on Azure (Lift and shift/Azure Migrate, other third-party tools) worked on Azure suite: **Azure SQL Database**, **Azure Data Lake (ADLS)**, **Azure Data Factory (ADF)** V2, Azure SQL Data Warehouse, Azure Service Bus, **Azure key Vault**, Azure Analysis Service (AAS), **Azure Blob Storage**, Azure Search, Azure App Service, Azure data Platform Services.
- Expert at data transformations like lookups, Derived Column, Conditional Splits, Sort, Data Conversation, Multicast and Derived columns, Union All, Merge Joins, Merge, Fuzzy Lookup, Fuzzy Grouping, Pivot, Un - pivot and SCD to load data in SQL SERVER destination.
- Proficient in Query/application tuning using optimizer Hints, explain plan, **SQL Trace**, **Index tuning** wizard, **SQL Profiler** and Windows Performance Monitor.
- Expertise in Developing Spark application using PySpark and Spark Streaming APIs in Python, deploying in **yarn cluster** in client, cluster mode.
- Hands on Experience on SSIS package deployment and scheduling.
- Import the data from different sources like HDFS/HBase into Spark RDD and perform computations using PySpark to generate the output response and configured Oozie workflows to generate Analytical Reports.

- Experience in creating various SSRS Reports like Charts, Filters, Sub-Reports, Scorecards, Drilldown and Drill-Through, Cascade, Parameterized reports that involved conditional formatting.
- Experience in report writing using SQL Server Reporting Services (SSRS) and creating several types of reports like Dynamic, Linked, Parameterized, Cascading, Conditional, Table, Matrix, Chart, Document Map and Sub-Reports.
- Experience developing iterative Algorithms using **Spark Streaming** in Scala and Python to build near real-time **dashboards**.
- Designing Enterprise reports using SQL Server Reporting Services and Excel Pivot table based on OLAP cubes which make use of multiple value selection in Parameters pick list, cascading prompts, matrix **dynamics reports** and other features of reporting service.
- Defect and story tracking using **JIRA**.

TECHNICAL SKILLS:

Azure Cloud Platform	Azure Data Factory v2, Azure Blob Storage, Azure Data Lake Gen 1 & Gen 2, Azure SQL DB, SQL server, Logic Apps.
Hadoop Distributions	Apache Hadoop 1x/2x, Cloudera CDP, Hortonworks HDP.
Languages	Python, Scala, Java, Pig Latin, HiveQL, Shell Scripting.
Software Methodologies	Agile, SDLC Waterfall.
Databases	MS SQL Server 2016-2008 R2, Oracle 12c, My SQL, MS Access, NoSQL DB: MongoDB, Azure SQL Datawarehouse.
NoSQL	HBase, MongoDB, Cassandra.
ETL/BI	Power BI, Tableau, Informatica.
Version control	GIT, SVN, Bitbucket.
Operating Systems	Windows (XP/7/8/10), Linux (Unix, Ubuntu), Mac OS.

PROFESSIONAL EXPERIENCE:

Client: Philips – Andover, MA

Mar 2022 - Present

Role: Azure Data Engineer

Responsibilities:

- Design and implement end-to-end data solutions (storage, integration, processing, visualization) in Azure.
- Implement ETL and data movement solutions using Azure Data Factory, SSIS.
- Develop dashboards and visualizations to help business users analyze data as well as providing data insight to upper management with a focus on Microsoft products like SQL Server Reporting Services (SSRS) and Power BI.

- Designed and developed enterprise scale cloud alert mechanism using azure Databricks, Spark/Spark UI data processing framework (Python/Scala) and azure Data Factory. Built data pipelines to transform, aggregate and process data using Azure Databricks, Azure ADLS, Blob, Azure Delta, and Airflow.
- Developed Databricks ETL pipelines using notebooks, Spark Data frames, SPARK SQL, and python scripting.
- Migrate data from traditional database systems to Azure SQL databases.
- Implement ad-hoc analysis solutions using Azure Data Lake Analytics/Store, HDInsight.
- Design and implement streaming solutions using Kafka or Azure Stream Analytics.
- Implemented near real-time data importing into Hadoop using Kafka and scheduled Oozie jobs.
- Experience managing Azure Data Lakes (ADLS) and Data Lake Analytics and an understanding of how to integrate with other Azure Services. Knowledge of USQL and how it can be used for data transformation as part of a cloud data integration strategy.
- Work with similar Microsoft on-prem data platforms, specifically SQL Server and related technologies such as SSIS, SSRS, and SSAS.
- Created Data Warehousing solutions using AWS Redshift from time to time according to client flexibility requirements.
- Developed Spark applications using PySpark and Spark-SQL for data extraction, transformation, and aggregation from multiple file formats for analyzing & transforming the data to uncover insights into the customer usage patterns.
- Recreating existing application logic and functionality in the Azure Data Lake, Data Factory, SQL Database and SQL Datawarehouse environment.
- Experience in DWH/BI project implementation using Azure DF.
- Involved in designing Logical and Physical Data Model for Staging, DWH and Data Mart layer.
- Created POWER BI Visualizations and Dashboards as per the requirements.
- Used various sources to pull data into Power BI such as SQL Server, Excel, Oracle, SQL Azure etc.
- Develop dashboards and visualizations to help business users analyze data as well as providing data insight to upper management with a focus on Microsoft products like SQL Server Reporting Services (SSRS) and Power BI.
- Engage with business users to gather requirements, design visualizations, and provide training to use self-service BI tools.
- Participating in Technical Architecture Documents, Project Design, and Implementation Discussions
- Azure Automation through Runbooks Creation, Migration of existing .PS1 scripts, Authorizing, Configuring, Scheduling.

Environment: Azure SQL, Azure Data Factory, Azure Storage Explorer, Azure Blob, Power BI Desktop, PowerShell, C#, .Net, Adobe Analytics, Fiddler, SSIS, SSAS, SSRS, DataGrid, Extract Transformation and Load (ETL), Business Intelligence (BI), Azure Storage, Azure Blob Storage, Azure Backup, Azure Files, Azure Data Lake Storage Gen 1/Gen 2, PySpark.

Aflac – Columbus, GA
Sr Azure Data Engineer

Apr 2020 – Feb 2022

Responsibilities:

- Create and maintain reporting infrastructure to facilitate visual representation of manufacturing data for purposes of operations planning and execution.
- Extract, Transform and Load data from source systems to Azure Data Storage services using a combination of Azure Data Factory, T-SQL, Spark SQL and Azure Data Lake Analytics.
- Implemented Restful web service to interact with Redis Cache framework.

- Intake happens through Sqoop and Ingestion happens through Map Reduce, HBASE.
- Extensively worked on Spark Streaming and Apache Kafka to fetch live stream data.
- Responsible for performing Machine-learning techniques regression/classification to predict the outcomes.
- Constructed product-usage SDK data and data aggregations by using PySpark, Scala,
- Spark SQL and Hive context in partitioned Hive external tables maintained in AWS S3 location for reporting, data science dashboarding, and ad-hoc analyses.
- Involved in data processing using an ETL pipeline orchestrated by AWS Data Pipeline using Hive.
- Installed Kafka manager for consumer lags and for monitoring Kafka Metrics also this has been used for adding topics, Partitions etc.
- Experience in creating configuration files to deploy the SSIS packages across all environments.
- Experience in writing queries in SQL and R to extract, transform and load (ETL) data from large data sets using Data Staging.
- Implemented CI/CD pipelines using Jenkins and built and deployed the applications.
- Worked on developing Restful endpoints to cache application specific data in in-memory data clusters like Redis and exposed them with Restful endpoints.
- Creating Databricks notebooks using SQL, Python and automated notebooks using jobs.
- Interacting with other data scientists and architected custom solutions for data visualization using tools like Tableau, packages in R.
- Developed predictive models using Python & R to predict customers churn and classification of customers.
- Documenting the best practices and target approach for CI/CD pipeline.
- Coordinated with QA team in preparing for compatibility testing of Guidewire solution.
- Familiar with data architecture including data ingestion pipeline design, Hadoop information architecture, data modelling and data mining, machine learning and advanced data processing.
- Designed and implemented by configuring Topics in the new Kafka cluster in all environments.

Environment: Hadoop, ETL operations, Data Warehousing, Data Modelling, Cassandra, AWS Cloud computing architecture, EC2, S3, Advanced SQL methods, NiFi, Python, Linux, Apache Spark, Scala, Spark-SQL, HBase

Nutanix - San Jose, CA

Nov 2018 – Mar 2020

Role: Azure Data Engineer

Responsibilities:

- Following Agile (Scrum) Methodology for developing application development.
- Design & implement migration strategies for traditional systems on Azure (Lift and shift/Azure Migrate, other third-party tools).
- Extract Transform and Load data from Sources Systems to Azure Data Storage services using a combination of Azure Data Factory, T-SQL, Spark SQL, and U-SQL Azure Data Lake Analytics. Data Ingestion to one or more Azure Services - (Azure Data Lake, Azure Storage, Azure SQL, Azure DW) and processing the data in In Azure Databricks.
- Worked on Ingesting data by going through cleansing and transformations and leveraging AWS Lambda, AWS Glue

- Worked on migration of data from On-prem SQL server to Cloud databases (Azure Synapse Analytics (DW) & Azure SQL DB).
- Recreating existing application logic and functionality in the Azure Data Lake, Data Factory, SQL Database and SQL data warehouse environment. Experience in DWH/BI project implementation using Azure DF.
- Created Pipelines in ADF using Linked Services/Datasets/Pipeline/ to Extract, Transform, and load data from different sources like Azure SQL, Blob storage, Azure SQL Data warehouse, write-back tool and backwards.
- Developed Spark applications using PySpark and Spark-SQL for data extraction, transformation, and aggregation from multiple file formats for analyzing & transforming the data to uncover insights into the customer usage patterns.
- Transformed data using AWS Glue dynamic frames with PySpark; cataloged the transformed the data using Crawlers and scheduled the job and crawler using workflow feature.
- Responsible for estimating the cluster size, monitoring, and troubleshooting of the Spark Databricks cluster.
- Experienced in performance tuning of Spark Applications for setting right Batch Interval time, correct level of Parallelism and memory tuning.
- Transformed data using AWS Glue dynamic frames with PySpark; cataloged the transformed the data using Crawlers and scheduled the job and crawler using workflow feature.
- Monitoring end to end integration using Azure monitor.
- Created Build and Release for multiple projects (modules) in production environment using Visual Studio Team Services (VSTS).
- Created POWER BI Visualizations and Dashboards as per the requirements.
- Used various sources to pull data into Power BI such as SQL Server, Excel, Oracle, SQL Azure etc.

Environment: Python, SQL, Oracle, Hive, Scala, Power BI, Azure Data Factory, Data Lake, Docker, MongoDB, Kubernetes, PySpark, SNS, Kafka, Data Warehouse, Sqoop, Pig, Zookeeper, Flume, Hadoop, Airflow, Spark, EMR, EC2, S3, Git, GCP, Lambda, Glue, ETL, Databricks, Snowflake.

Client: Cloud Digital Media – India

Sept 2017 – Sept 2018

Role: Azure Data Engineer

Responsibilities:

- Participate with technical staff team and business managers and practitioners in the business unit to determine requirements and functionalities needed in a project.
- Performed wide, narrow transformations, actions like filter, Lookup, Join, count, etc. on **Spark Data Frames**.
- Worked with Parquet files and Json using **Spark**, and **Spark Streaming** with Data Frames.
- Developed batch and streaming processing apps using Spark APIs for functional pipeline requirements.

- Created **Spark** application that uses **Spark SQL** to generate data frames from Avro formatted raw layer and writes them to data service layer internal tables as Parquet format.
- Experienced import/export data into **S3/Hive** from relational database and Teradata using **Spark JDBC**.
- Involved in the creation of **Hive** tables, loading, and analyzing the data by using hive queries.
- Have worked on creating and configuration of **EMR** clusters on **AWS** cloud for running spark workloads.
- Worked on **CI/CD** solution, using **Git, Jenkins, Docker** to setup and configure big data architecture on **AWS cloud** platform.
- Experienced in writing **Spark Applications** in Scala and Python (PySpark).
- Implement **Spark** applications using python to perform advanced procedures like text analytics and processing, utilizing data frames and **Spark SQL API** with in-memory computing capabilities of **Spark** for faster processing of data.
- Wrote **Spark-Streaming** applications to consume the data from **Kafka** topics and write the processed streams to **Redshift**.
- Worked on **AWS Lambda Functions** to run serverless functions for file validation and file copy and to trigger step functions.
- Working on integrating Kafka Publisher in spark job to capture errors from **Spark** Application and push into database.

Environment: AWS EMR, Spark, Hive, Kafka, UNIX, Shell, AWS Services, Python, Scala, Glue, SQL.

Client: Info Softech – India

Feb 2015 – Aug 2017

Role: Data Engineer

Responsibilities:

- Interacted with business partners, Business Analysts and product owners to understand requirements and build scalable distributed data solutions using the Hadoop ecosystem.
- Developed Spark Streaming programs to process near real time data from Kafka, and process data with both **stateless and Stateful transformations**.
- Worked with **HIVE** data warehouse infrastructure-creating tables, data distribution by implementing **partitioning and bucketing**, writing, and optimizing the **HQL queries**.
- Built and implemented automated procedures to split large files into smaller batches of data to facilitate FTP transfer which reduced 60% of execution time.
- Written multiple MapReduce Jobs using **Java API**, Pig and Hive for data extraction, transformation and aggregation from multiple file formats including Parquet, Avro, XML, JSON, CSV, ORCFILE and other compressed file formats Codecs like gZip, Snappy, Lzo.
- Strong understanding of Partitioning, bucketing concepts in Hive and designed both Managed and External tables in Hive to optimize performance.

- Developed PIG UDFs for manipulating the data according to Business Requirements and worked on developing custom PIG Loaders.
- Developing ETL pipelines in and out of data warehouses using a combination of Python and **Snowflakes SnowSQL** Writing SQL queries against Snowflake.
- Worked on installing cluster, commissioning & decommissioning of data node, name node recovery, capacity planning, and slots configuration.
- Developed data pipeline programs with Spark Scala APIs, data aggregations with Hive, and formatting data (JSON) for visualization.

Environment: Cassandra, PySpark, Apache Spark, HBase, Apache Kafka, HIVE, SQOOP, FLUME, Apache oozie, Zookeeper, ETL, UDF, Map Reduce, Snowflake, Apache Pig, Python, Java, SSRS.